# PWR 初学者教学版 (中文) v20261017-0755

## ① 一句话

Quanta Services, Inc. 是一家 Power grid / utility infrastructure 电网基建 公司。现在股价 $631.86。我们的 Base 模型认为每股价值约 $229（-63.7%）。

## ② 发生了什么

最近一季财报 EPS 实际 vs 预期: [‘2026-06-30’, ‘4.24’, ‘3.30957’, ‘0.2811’]（实际、预期、超预期%）。下次财报约 2026-10-29。

## ③ 术语 Glossary

- DCF (Discounted Cash Flow｜现金流折现): 把未来每年能拿到的现金，按”时间成本”折回今天加总。
- FCF (Free Cash Flow｜自由现金流) = 经营现金流 − 资本开支。
- WACC (Weighted Average Cost of Capital｜加权平均资本成本): 投资人要求的年回报率，用作折现率。
- Reverse DCF (反向 DCF): 反过来问”现价要求公司增长多快”。
- Consensus (一致预期): 分析师预测的平均值。
- Falsifier (证伪指标): 一旦出现就说明论点错了的信号。

## ④ 生活比喻

买这家公司股票≈买一棵果树。DCF 就是估算果树未来每年结多少果(FCF)，再考虑”明年的一斤果不如今天的一斤果值钱”(WACC 折现)。

## ⑤ 因果箭头

Power grid / utility infrastructure 电网基建 需求 → 收入增长 (g) → 自由现金流 (FCF = 收入 × FCF 率) → 折现 (÷(1+WACC)^t) → 每股价值 → 与股价比较 → 预期差

## ⑥ 数字例题（符号定义）

设 $R_0$ = 过去 12 个月收入 = 32.9 bn USD；$g_1$ = 第一年增速 = 15.8%；m = 长期 FCF 率 = 6.0%；W = WACC = 11.4%。 第一年收入 $R_1 = R_0 \times (1+g_1)$ = 38.1 bn USD。 若第 5 年 FCF 率达到 m，则第 1 年 FCF ≈ $R_1 \times m_1$（$m_1$ 由当前 7.3% 向 6.0% 线性过渡）= 2.7

bn USD； 折现到今天: $FCF_1 \div (1+W)^1$ = 2.4 bn USD。十年加总 + 终值 = EV 40.6 bn USD。

## ⑦ 事实 / 预测 / 市场预期

- 事实: 过去财务数据(第 4 节)、现价。
- 预测(我们的假设): 长期 FCF 率 6.0%、三情景。
- 市场预期: 一致预期增速 15.8%；价格隐含增速 40.9%。

## ⑧ 三情景

Bear $103 / Base $229 / Bull $372；概率加权 $233。

## ⑨ 反方观点与失效条件

反方: Permitting delays; renewable policy reversal｜审批延迟；可再生政策逆转 失效条件: Backlog growth stalls; utility capex cut｜积压停滞；电力 Capex 下修

## 自我检验题

1. 如果 WACC 上升 1 个百分点，价值会升还是降？为什么？（提示：分母变大）
2. 反向 DCF 隐含增速高于一致预期意味着什么？

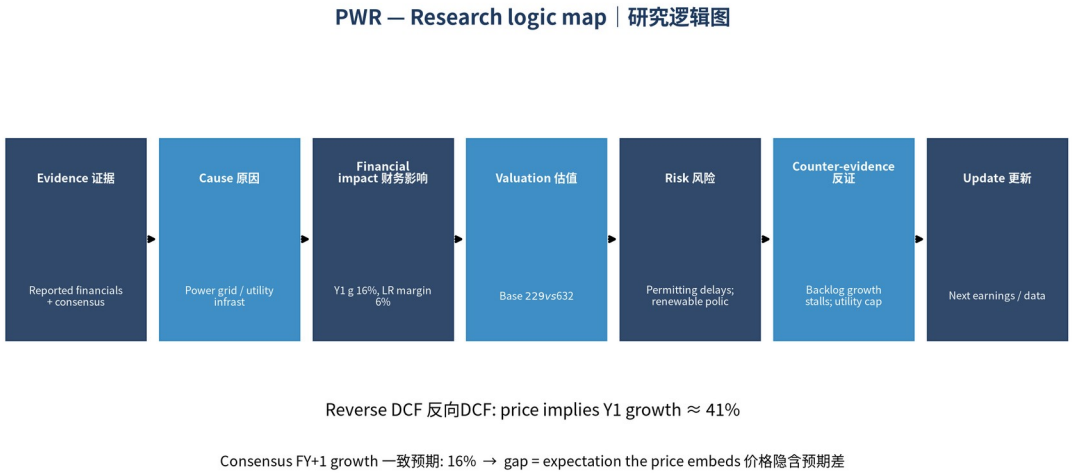
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# PWR Beginner Teaching Edition (English) v20261017-0755

## ① In one line

Quanta Services, Inc. is a Power grid / utility infrastructure 电网基建 company. Price $631.86; our Base model value ≈ $229 (-63.7%). ## ② What happened Last quarter EPS actual vs consensus: [‘2026-06-30’, ‘4.24’, ‘3.30957’, ‘0.2811’]. Next earnings ≈ 2026-10-29 (TENTATIVE). ## ③ Glossary DCF (Discounted Cash Flow｜现金流折现); FCF (Free Cash Flow｜自由现金流) = operating cash flow − capex; WACC (Weighted Average Cost of Capital｜加权平均资本成本) = the yearly return investors demand; Reverse DCF (反向 DCF) = what growth the price requires; Consensus (一致预期); Falsifier (证伪指标). ## ④ Analogy Owning the stock ≈ owning a fruit tree: DCF estimates future fruit (FCF) and discounts it because fruit next year is worth less than fruit today. ## ⑤ Causal arrows Demand → revenue growth (g) → FCF (= revenue × FCF margin) → discounting ÷(1+WACC)^t → value/share → compare with price → expectation gap

## ⑥ Worked example (symbols) $R_0$ = TTM revenue = 32.9 bn USD; $g_1$ = 15.8%; m = long-run FCF margin = 6.0%; W = 11.4%. $R_1 = R_0 \times (1+g_1)$ = 38.1 bn USD; $FCF_1$ = 2.7 bn USD; PV = $FCF_1/(1+W)$ = 2.4 bn USD; 10 years + terminal → EV 40.6 bn USD. ## ⑦ Fact vs forecast vs market expectation Facts: reported financials, price. Forecast: our margin/scenario assumptions. Market: consensus growth 15.8%; price-implied growth 40.9%. ## ⑧ Three scenarios Bear $103 / Base $229 / Bull $372; weighted $233. ## ⑨ Counterview & failure conditions Permitting delays; renewable policy reversal｜审批延迟；可再生政策逆转 — Falsifiers: Backlog growth stalls; utility capex cut｜积压停滞；电力 Capex 下修 ## Self-check 1. WACC +1pp → value up or down? 2. What does price-implied growth > consensus mean?




Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

*logic*